

# Case Study: Mathematical Modelling

Brownian Microswimmers

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# 1 Introduction

Consider a continuous random variable  $X$  which has been ‘independently and identically distributed’ (i.i.d.), from the uniform density distribution  $\mathcal{U}(0, 2\pi)$ . We seek to find the probability density function of  $X$ , with a probability density function  $f(x)$ , given by:

$$f(x) = \begin{cases} \frac{1}{2\pi} & \text{if } x \in [0, 2\pi] \\ 0 & \text{otherwise.} \end{cases}$$

To do this, let us define  $m$  bins of a histogram by  $\Delta = \frac{2\pi}{m}$ , so that the interval of the  $i^{\text{th}}$  bin  $B_i$  is given by  $B_i = [(i-1)\Delta, i\Delta)$ , for  $i = (1, \dots, m)$ . Let us now take  $n$  random samples, of  $X$ , and denote them  $X_1, \dots, X_n$ . Then the normalised height over bin  $B_i$  is

$$H_n(x_i) = \frac{1}{\Delta n} \sum_{j=1}^n \mathbb{1}_{\{X_j \in B_i\}},$$

where we have taken  $x_i$  to be the midpoint of  $B_i$ , and  $\mathbb{1}$  to be the standard indicator function. Then, our expectation of the height is given by

$$\mathbb{P}(X_j \in B_i) = \int_{B_i} f(x) dx = \Delta \cdot \frac{1}{2\pi},$$

and therefore we can find the expectation value to be

$$\mathbb{E}[H_n(x_i)] = \frac{1}{\Delta n} \sum_{j=1}^n \mathbb{P}(X_j \in B_i) = \frac{1}{2\pi}.$$

In other words, every bin  $B_i$  has the same expected height, recovering a uniform density. Now, let us assume that  $X$  is the random variable corresponding to a direction in two-dimensions. We see that this would suggest objects moving randomly would have a uniform distribution of their alignments.

The term ‘Brownian microswimmers’ pertains to any micro-organism that is capable of self-propulsion and is observed to move through a fluid in random directions. Our motivation to study their movements arises because it is well documented (see Rothschild [11] and Lauga [8]) that although such microswimmers move randomly throughout their vessel, they are found to accumulate non-trivially at boundaries, rather than in the bulk of the medium in which they swim. In this Case Study, we aim to explain the physical reasons this occurs, by employing three models which demonstrate this phenomenon, and use these models in our own computations, before analysing our results and comparing the relative advantages and disadvantages of each model.

## 2 Theory

The three models we shall consider all have foundation in stochastic differential equations. This is sensible due to the innate randomness we see empirically in the behaviour of the microswimmers. A brief outline of each model is given in Table 1 below.

| Model   | Description                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------------------|
| Model 1 | A simple stochastic model where stochastic differential equations govern the motion and orientation of the microswimmer.            |
| Model 2 | A hydrodynamic model arising from the force dipoles generated by the microswimmer swimming, with stochastic corrections.            |
| Model 3 | A simplified stochastic model that differentiates between microswimmers near boundaries and microswimmers in the bulk of the fluid. |

Table 1: A brief summary of the three models used in this study. Throughout this report, we shall refer to them by their assigned model numbers.

It is clear that an understanding of stochastic differential equations is essential to the study of Brownian microswimmers, so we will introduce them here (see Erban and Chapman [4]). Consider the variable  $x(t) \in \mathbb{R}$ , the evolution of which is described by an ordinary differential equation  $\frac{dx}{dt} = f(x(t), t)$ , for some given function  $f : \mathbb{R} \times [0, \infty) \rightarrow \mathbb{R}$ . We can then rewrite this expression in its differential form through the definition of the derivative given by  $dx = f(x(t), t)dt$ . Intuitively, we see that the derivative  $dx$  describes the infinitesimal change in  $x$  over an infinitesimal change in  $t$ , otherwise written  $dx = x(t+dt) - x(t)$ . Combining these expressions, and substituting the differentials  $dx$  and  $dt$  for small (but no longer infinitesimal) intervals  $\Delta x$  and  $\Delta t$ , we can reformulate our ordinary differential equation into its *computational definition*:

$$x(t + \Delta t) = x(t) + f(x(t), t)\Delta t, \quad (1)$$

which can then be solved numerically using a finite difference scheme. In general terms, a stochastic differential equation can be thought of as adding a ‘white noise’ term to expression (1). Then, for our computational definition of a stochastic differential equation governing the time evolution of the random positional variable  $X(t)$ , we can write

$$X(t + \Delta t) = X(t) + f(X(t), t)\Delta t + g(X(t), t)\sqrt{\Delta t}\xi, \quad (2)$$

where  $f$  is defined as before,  $g : \mathbb{R} \times [0, \infty) \rightarrow \mathbb{R}$  is given and governs the relative strength or influence of the white noise, and  $\xi$  is our random variable, taken from

the standard Normal distribution,  $\xi \sim \mathcal{N}(0, 1)$ . It is common to define the combined terms  $\sqrt{\Delta t}\xi$  as a singular term  $\Delta W$ , known as the Wiener term, given by  $\Delta W \sim \mathcal{N}(0, \Delta t)$ . As a result, it is clear that a probabilistic approach to our numerical solutions over many iterations is more useful in describing the underlying dynamics of (2) than a singular trial.

Let us therefore define the probability distribution function  $p(x, t)$  of (2) such that  $p(x, t)dx$  describes the probability of that  $X(t) \in [x, x + dx]$ . One can show that the evolution of  $p(x, t)$  can be described by the Fokker-Planck equation (see Risken [10]).

**Theorem 2.1** (The Fokker-Planck equation). The evolution of the probability distribution function  $p(x, t)$  of the random variable  $X(t)$  governed by (2) is described by the Fokker-Planck equation:

$$\frac{\partial p(x, t)}{\partial t} = \frac{\partial^2}{\partial x^2} \left( \frac{g^2(x, t)}{2} p(x, t) \right) - \frac{\partial}{\partial x} (f(x, t)p(x, t)). \quad (3)$$

We have defined this equation for one-dimensional stochastic differential equations, although it can be generalised to higher dimensions.

We have one final concept that we must define before we may proceed to our three models, namely the *computational domain* in which we had defined our simulations. Numerous studies, such as Chen and Thiffeault [2], show that the shape of the microswimmers has a substantial impact on their distribution throughout their vessel. For our simulations, we must therefore find a way to incorporate the shape of the microswimmers and their interactions with the boundaries as a result into it. Model 1 and Model 2 rely on considering the microswimmer as a rigid ellipse, with this in mind we shall create the domain we see in Figure 1.

Once the computational domain has been defined, we can simulate our microswimmer as a point particle representing the centre of the ellipse, moving through it. Indeed, Figure 1 illustrates the most intuitive way to understand the computational domain. Given a horizontal surface, representing a side of the vessel the microswimmers are contained in, and the shape of the microswimmer we are considering, we can rotate the shape of the microswimmer without slipping along the edge. Taking the distance from the edge to the centre of the microswimmer as it rolls yields the computational domain in phase space. Each model we introduce shall only consider the orientation of the microswimmer and its vertical displacement, which we may denote as the coordinate system  $(y, \theta)$ .

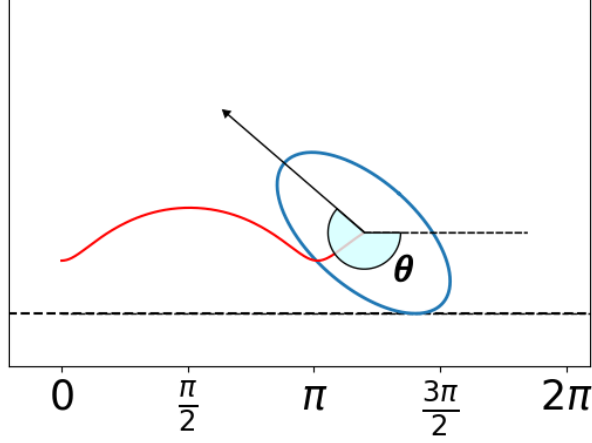
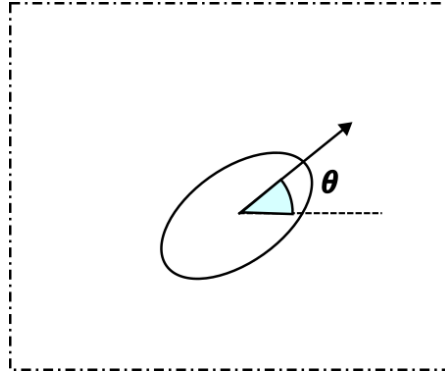
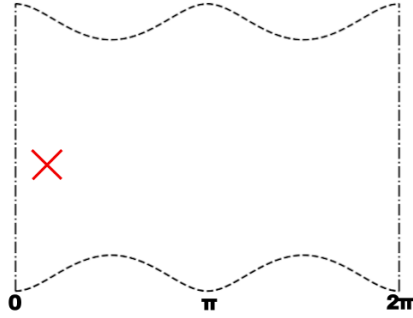


Figure 1: A phase plane with the horizontal axis given by the angle  $\theta$ , measured from the horizontal. The red line represents the vertical height of the centre of the ellipse as it rotates along the axis. This is how we form the computational domain.

We neglect the  $x$ -direction as it would have the same dynamics as the  $y$ -direction, and thus does not provide any additional insight. Furthermore, with an additional dimension, we would no longer be able to represent the full probability distribution function  $p(y, \theta)$  as a surface in three-dimensions.



(a) A diagram of a microswimmer in physical space.



(b) A diagram of the location of the microswimmer in the computational domain.

Figure 2: A comparison of a microswimmer orientation and position in physical space versus its position in the computational domain, considering only its distance from the horizontal axes.

## 2.1 Model 1

The first model we shall consider is an elliptic cell in a channel which is infinite across its length  $x$ , but with a width  $y = H$ . As a result, we can discard movement

in  $x$ . Since microswimmers by their definition exhibit motility, we shall also add a component of velocity  $U$  in the  $y$ -direction. As we want the microswimmer to swim in random directions, our orientation shall only be based upon the orientation in the previous increment and a stochastic diffusive term. Putting this together, we find for Model 1:

$$dY = U \sin(\theta)dt + \sqrt{2D_y} dW_1 \quad (4a)$$

$$d\Theta = \sqrt{2D_\theta} dW_2, \quad (4b)$$

where  $Y$  and  $\Theta$  represent the random variables, and we have defined  $dY := Y(t+dt) - Y(t)$ , and similarly for  $d\Theta$ . Values  $D_y$  and  $D_\theta$  represent the diffusion constants, and  $dW_1$  and  $dW_2$  representing Weiner differentials in the  $y$  and  $\theta$ -directions respectively. Note that we have used two additions of white noise here, such that  $dW_1 \sim \mathcal{N}_1(0, dt)$  and  $dW_2 \sim \mathcal{N}_2(0, dt)$ . This is because we must sample each random variable from different identical Normal distributions, so that there is no correlation between the step size and the orientation, which would introduce a further drift component. The resulting Fokker-Planck equation is given by:

$$\begin{aligned} \frac{\partial p}{\partial t} = & -\frac{\partial}{\partial y} (U \sin(\theta)p) + D_y \frac{\partial^2 p}{\partial y^2} + D_\theta \frac{\partial^2 p}{\partial \theta^2} \\ & \int_{\theta=0}^{\theta=2\pi} \int_{y=0}^{y=H} p(y, \theta, t) dy d\theta = 1. \end{aligned} \quad (5)$$

We must also consider how the cell interacts with the boundary when it moves. Numerous different models for boundary interactions have been considered in the present literature, such as a harmonic potential boundary used in Drescher *et al.* [3], or a no-flux boundary, as employed in Chen and Thiffeault [2]. We have chosen to use a reflective boundary condition, where once a point on the ellipse contacts the boundary, it is reflected by constructing the perpendicular bisector of the boundary and using this normal to return the reflected position. The steps are shown in Figure 3 below. Once the position of intersection is found, we calculate the numerical derivative denoted `dy_dtheta` using the central difference expression, to construct the tangent to the surface which is then normalised. We can then project along the tangential direction, before calculating the reflection of the initial point.

```

def find_intersection(p0, p1, boundary_function, a, b):
    theta0, theta1 = p0[0], p1[0]
    dtheta = theta1 - theta0
    y0, y1 = p0[1], p1[1]
    dy = y1 - y0
    def error(t):
        ...
    return collision_point

def give_reflection(p_intersect, p1, boundary_func, a, b):
    theta_intersect = p_intersect[0]
    dy_dtheta = numerical_derivative(a, b, theta_intersect, boundary_func)[0]
    tangent = np.array([1, dy_dtheta]).reshape((2,1))
    tangent = tangent/np.linalg.norm(tangent)
    projection = p_intersect + np.dot((p1-p_intersect).T, tangent) * tangent
    reflection = 2 * projection - p1
    return reflection

```

Figure 3: A snippet of the code we used in Model 1, showing how we implemented the reflective boundary conditions. The function `error(t)` has been omitted for brevity, and solves for the cell’s intersection with the boundary using an optimised classical bisection algorithm, from the `scipy.optimize` library.

We had also enforced periodic boundary conditions for  $\theta$ , namely that `theta_next = theta_next % (2 * np.pi)` in our code, in order for the orientation of the cell to remain continuous.

## 2.2 Model 2

Strictly speaking, we shall consider two sub-models that use hydrodynamic interactions, in order to explain the boundary accumulation of microswimmers. The first model we shall introduce is what we shall call the ‘pure hydrodynamic model’, such as the one introduced in Berke *et al.* [1] and Spagnolie and Lauga [12]. This model attributes the cell distribution profile entirely to hydrodynamic effects, caused by the resulting Stokes’ equations of force dipoles generated by a swimming micro-organisms. The second model we shall consider will be another stochastic model, which uses additional hydrodynamic terms in order to capture these fluid effects. This model was demonstrated in two-dimensions in §8 of Chen and Thiffeault [2].

Before we begin describing each model, we must now introduce some definitions pertaining to the motility of our microswimmers. A *pusher* is a micro-organism that has its flagellum or flagella at the rear of its body and provides propulsion usually through chiral rotation, as described in Lauga [8]. A common example of a pusher

used in the literature is *Escherichia coli*, and a simple schematic of a general pusher is found in Figure 4a.

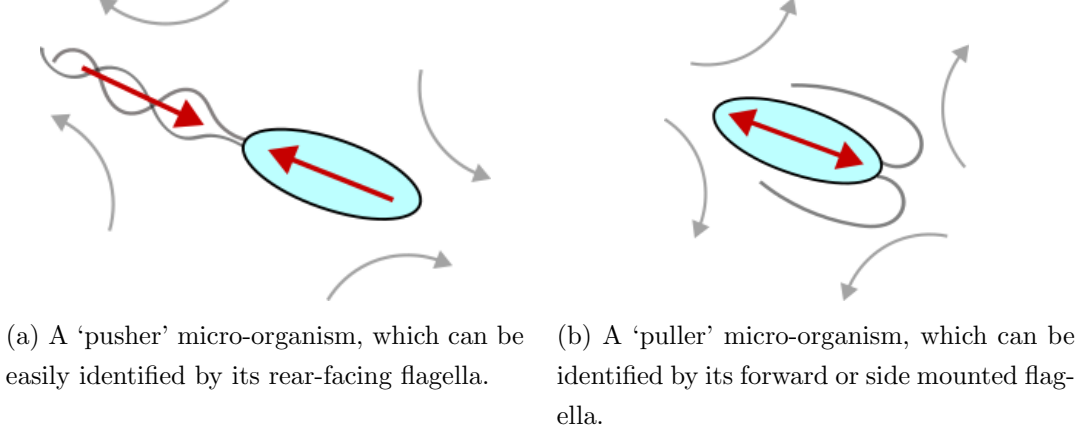


Figure 4: A comparison of ‘pusher’ and ‘puller’ micro-organisms, with red arrows denoting the dipole forces and grey arrows denoting fluid streamlines.

A *puller* is a micro-organism which depends on forward-facing flagella which provide motility. We can see in Figure 4 that each have different resulting force dipoles and streamlines arising from their motion, which will impact the mechanisms of their boundary accumulation. In our studies, we mainly considered pushers.

It is useful to note that most hydrodynamics models considered use Stokes’ equations in order to model the velocity fields surrounding the micro-organism. This is because the Reynold number,  $Re$ , is typically less than  $10^{-2}$  in the resulting situation, as demonstrated in Pedley and Kessler [9], where slow viscous flow dominates. We can think of the force dipole as two point forces, and solve the resulting system. Upon doing so, we find the *stokeslet*, a singular solution of Stokes’ equations. This in turn can be decomposed into a symmetric *stresslet* and an antisymmetric *rotlet*. As we assume our force dipole is symmetric (we do not expect the microswimmer to be meaningfully accelerating instantaneously), we only recover the symmetric streamlines of the stresslet. In our first model, employing this procedure, we find that the induced  $y$ -component of velocity in the direction away from the no-slip boundary is given by

$$u_y(y, \theta) = -\frac{3P}{64\pi\eta y^2}(1 - 3\sin^2(\theta)), \quad (6)$$

where  $P$  denotes the dipole moment and  $\eta$  denotes the viscosity. We may also use the derivation in Jeffery [6] in order to find that the rotation rate in the plane, denoted

$$\Omega_z(y, \theta) = -\frac{3P \sin(\theta) \cos(\theta)}{64\pi\eta y^3} \left[ 1 + \frac{\gamma^2 - 1}{2(\gamma^2 + 1)}(1 + \sin^2(\theta)) \right] \quad (7)$$

where we have  $\gamma$  as the aspect ratio of our ellipse. In the two-dimensional case, this is given by the ratio of the major axis to the minor axis,  $\frac{a}{b}$ . For pushers, such as *E. coli*,  $P > 0$ . We also have that the aspect ratio for most microswimmers is  $\gamma > 1$ , especially so if we take into consideration the flagella. As a result, the sign of  $\Omega_z(\theta, y)$  depends only on the sign of  $-\sin(\theta)\cos(\theta)$ . Then, we find for  $0 \leq \theta \leq \frac{\pi}{2}$ , the rotation rate is negative and hence the cell is reoriented parallel to the boundary. Similarly, for  $\frac{\pi}{2} < \theta \leq \pi$ , the rotation rate is positive and the cell is oriented parallel to the boundary, in the opposite direction. This model permits an analytical form of the steady state Fokker-Planck equation. Indeed, we have

$$\frac{d(pu_y)}{dy} = D_y \frac{\partial^2 p}{\partial y^2} \quad (8)$$

Then, for cells parallel to our two boundaries separated by a distance  $H$ , we have that  $u_y(y, \frac{\pi}{2}) = -\frac{3P}{64\pi\eta}(\frac{1}{y^2} - \frac{1}{(H-y)^2})$ , where we have neglected the higher order terms in  $\frac{1}{y}$  and  $\frac{1}{H}$  due to the reflections of each image system onto the boundaries. Now, let us assume that  $\frac{\partial p}{\partial y} = 0$  at  $y = \frac{H}{2}$ . Then, we can integrate (8) to find the number density distribution

$$\frac{p(y)}{p_0} = \exp \left[ L_y \left( \frac{1}{y} + \frac{1}{H-y} \right) \right], \quad L_y = \frac{3P}{64\pi\eta D_y}, \quad (9)$$

where we can fit constants  $p_0$  and  $L_y$  in accordance with observations. We see from the expression for  $p(y)$  that this is not a probability in the strict sense, as we must also normalise it so that the probability across the entire domain integrates to unity. Equation (9) has singularities at  $y = 0$  and  $y = H$ , arising from the derivation of the velocity field. This is because we had used a ‘far-field’ geometric approximation, so our solution becomes less accurate as we move closer to each edge.

The second hydrodynamic model we consider can be found in Chen and Thiffeault [2] and Spagnolie *et al.* [13]. In this case, we have the same stochastic differential equation as described in Model 1, but we add an additional hydrodynamic correction, yielding the expressions

$$dY = U \sin(\theta)dt - \frac{3PU}{8} \left( \frac{a^2}{y^2} - \frac{a^2}{(H-y)^2} \right) [1 - 3\sin^2(\theta)]dt + \sqrt{2D_y}dW_1 \quad (10a)$$

$$d\Theta = -\frac{3PU}{64a} \left( \frac{a^3}{y^3} + \frac{a^3}{(H-y)^3} \right) [4 - \Gamma(3 - \cos(2\theta))] \sin(2\theta)dt + \sqrt{2D_\theta}dW_2. \quad (10b)$$

We see that the first and third terms in (10a) are identical to the right-hand side of (4a), and the second term in (10b) is identical to the right-hand side of (4b), and we have defined  $\Gamma = \frac{a^2-b^2}{a^2+b^2} \in [0, 1]$ . Our method to solve this system remains largely the

same as the one described in Model 1, with the main difference in our code being the definitions of the numerical steps  $dY$  and  $d\Theta$ , pertaining to our new relations described above.

### 2.3 Model 3

In this section we present another stochastic model introduced in Fu *et al.* [5] that uses a different approach to boundary accumulation. We distinguish microswimmers according to whether they are in the ‘free-swimming’ phase or the ‘boundary-contacting’ phase. We assume the same rotational diffusion process as considered in Model 1. When a microswimmer hits the boundary, it moves from the free-swimming phase to the boundary-contacting phase. It does not move from this phase until the microswimmer is oriented away from the boundary towards the rest of the bulk medium. Model 3 is known as a Piecewise-Deterministic Markov Process, as its behaviour is governed by stochastic motion at jumps in time, but its motion is governed deterministically in between their jumps. The equations of motion for Model 3 are:

$$dY = \begin{cases} U \sin(\theta) dt, & \text{if } |Y| < \frac{H}{2} \text{ or } (|Y| = \frac{H}{2} \text{ and } Y\Theta \leq 0) \\ 0, & \text{if } |Y| = \frac{H}{2} \text{ and } Y\Theta > 0 \end{cases} \quad (11a)$$

$$d\Theta = \sqrt{2D_\theta} dW. \quad (11b)$$

Notice that we only have one Brownian process in these equations, in order to simplify the velocity of the microswimmers to an entirely deterministic expression. Equations (11) are defined in the union of sets  $\Omega \cup \Omega_+ \cup \Omega_-$ , where  $\Omega$  is the free-swimming phase given by  $\Omega := \{(y, \theta) \mid -\frac{H}{2} < y < \frac{H}{2}, -\pi < \theta < \pi\}$ , and  $\Omega_+$  and  $\Omega_-$  represent the top and bottom boundary-contacting phases respectively, and we can define them by  $\Omega_\pm := \{(y, \theta) \mid y = \pm \frac{H}{2}, 0 < \pm\theta < \pi\}$ .

We can also define the corresponding degenerate Fokker-Planck system, where we shall define  $p$  as before,  $p_+$  as the probability of the microswimmer in the upper boundary-contacting phase and  $p_-$  as the probability of the microswimmer in the lower boundary-contacting phase. Together, these yield the system:

$$\frac{\partial p_+}{\partial t} - D_\theta \frac{\partial^2 p_+}{\partial \theta^2} = U \sin(\theta) p(t, \frac{H}{2}, \theta) \quad (12a)$$

$$\frac{\partial p}{\partial t} + U \sin(\theta) \frac{\partial p}{\partial y} - D_\theta \frac{\partial^2 p}{\partial \theta^2} = 0 \quad (12b)$$

$$\frac{\partial p_-}{\partial t} - D_\theta \frac{\partial^2 p_-}{\partial \theta^2} = -U \sin(\theta) p(t, -\frac{H}{2}, \theta). \quad (12c)$$

We have yet to consider any boundary conditions to impose on the system. This is because we can no longer trivially close the system by using zero-flux boundary con-

ditions or reflective boundary conditions, as considered in Model 1 and 2. Our first problem arises since we have multiple probability distributions,  $p$ ,  $p_+$  and  $p_-$ . Therefore, such boundary conditions would have to consider the microswimmers moving between states governed by different dynamics. As a result, our boundary conditions would need to carefully review the flux terms for each boundary of the degenerate system to account for the exchange of microswimmers across the phases.

We must also ensure that we adhere to the conservation of mass probability, in particular  $\int_{\Omega} p dy d\theta + \int_{\Omega_-} p_- dy d\theta + \int_{\Omega_+} p_+ dy d\theta = 1$ . In order to address these difficulties, we shall employ the following boundary conditions:

$$p(t, y, -\pi) = p(t, y, \pi), \quad \frac{\partial p}{\partial \theta}(t, y, \pi) = \frac{\partial p}{\partial \theta}(t, y, -\pi) \quad (13a)$$

$$p(t, -\frac{H}{2}, \theta) = 0, \quad \theta \in (0, \pi) \quad (13b)$$

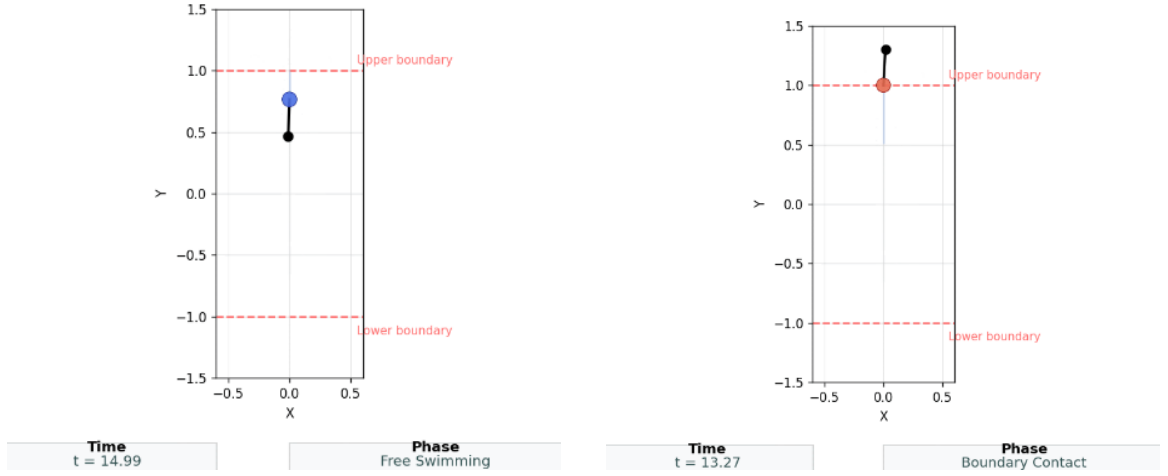
$$p(t, \frac{H}{2}, \theta) = 0, \quad \theta \in (-\pi, 0) \quad (13c)$$

$$p_+(t, 0) = p_+(t, \pi) = 0, \quad p_-(t, 0) = p_-(t, -\pi) = 0 \quad (13d)$$

$$\left( \frac{\partial p}{\partial \theta} \right) \Big|_{\theta=\pi_-}^{\pi_+} = \frac{\partial p_+}{\partial \theta} \delta(y - \frac{H}{2}) - \frac{\partial p_-}{\partial \theta} \delta(y + \frac{H}{2}) \quad (13e)$$

$$\left( \frac{\partial p}{\partial \theta} \right) \Big|_{\theta=0_-}^{0_+} = -\frac{\partial p_+}{\partial \theta} \delta(y - \frac{H}{2}) + \frac{\partial p_-}{\partial \theta} \delta(y + \frac{H}{2}). \quad (13f)$$

We have that the expressions in (13a) represent the periodicity in  $\theta$ . Conditions (13b) and (13c) ensure that that no microswimmer can enter from outside of the domain, as this would be unphysical. Conditions (13d) ensure that our microswimmer models cannot leave the boundary-contacting phases if they are pointing away from them. Finally, conditions (13e) and (13f) maintain the overall probability conservation for microswimmers that are switching dynamics between the free-swimming phase and the boundary-contacting phases, with a point source in the form of a Dirac  $\delta$ -function located at the points where the two phases intersect.



(a) The simplified microswimmer in the free-swimming phase.

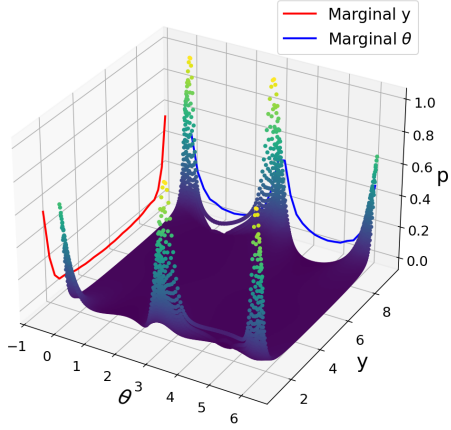
(b) The simplified microswimmer in the boundary-contacting phase.

Figure 5: A schematic to show the differing phases of Model 3. The black ball and stick showing the orientation direction. We visually distinguish the phases by their position and the colour of the microswimmer; blue when in the free-swimming phase and red when in the boundary-contacting phase.

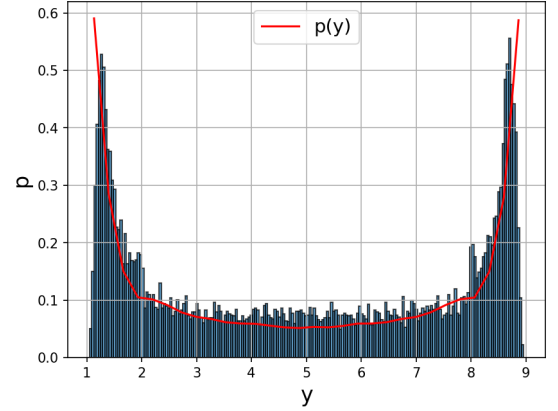
Figure 5 shows a schematic to distinguish the different phases of the model. For Model 3, we had used two different methods in order to calculate the probability distributions  $p$ ,  $p_-$  and  $p_+$ . In Fu *et al.* [5], the authors describe two methods to obtain these distributions, namely using a finite difference scheme to solve system (11) along with (12) and (13). The second method used Monte Carlo simulations as we had employed in Model 1 and Model 2.

### 3 Results

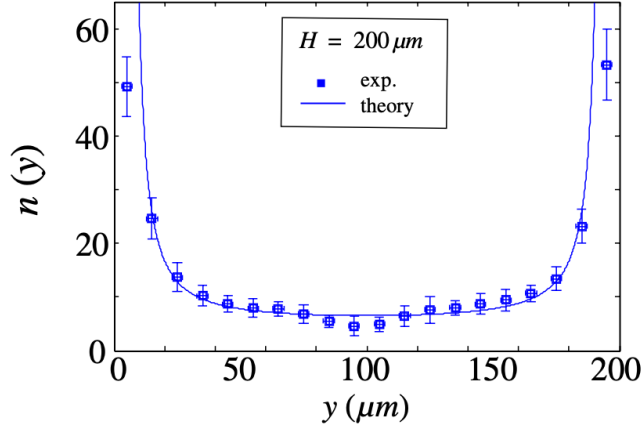
In broad terms, we can say that each model was effective in showing the increased probability of microswimmers accumulating on or near the boundaries, with similar time-invariant probability distributions exhibited by solving each model's corresponding Fokker-Planck equation either numerically or analytically, our numerical Monte Carlo simulations of each model, and the finite difference scheme suggested in Model 3. We shall examine the results from each model in turn, before analysing their relative efficacies.



(a) A histogram generated using the COMSOL Multiphysics software package in showing the numerical solution to (4) in Model 1 and the corresponding Fokker-Planck equations (5).



(b) A histogram generated through Monte Carlo simulations of equation (4a) and the overlaid marginal  $p(y)$  found in Figure 6a. We see a good agreement between the two methods.

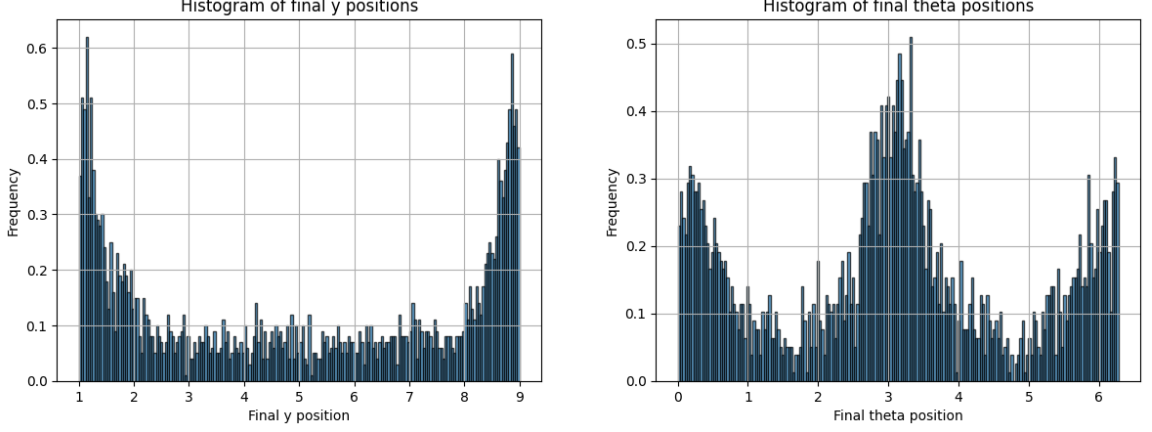


(c) Taken from Berke *et al.* [1]. We are particularly interested in the experimental results illustrated in this paper, shown as blue points above.

Figure 6: Three plots illustrating the boundary accumulation of microswimmers. Figure 6a and Figure 6b are derived from Model 1, whereas Figure 6c shows experimental results.

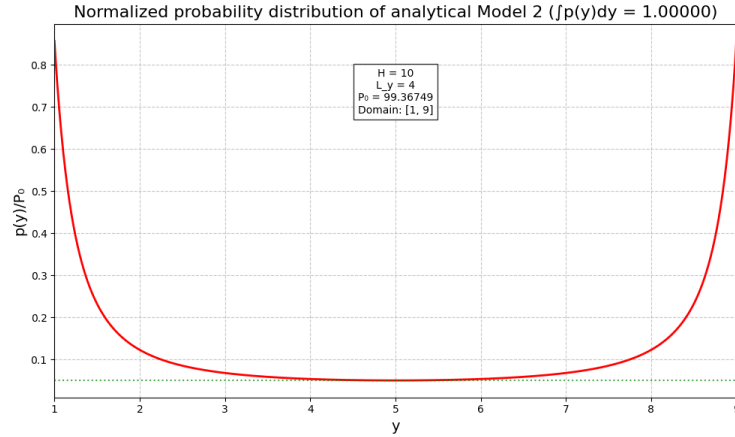
For the simulation of Model 1, we had used a vessel width of  $H = 10$ , and positioned the microswimmer in the centre of the computational domain at  $(y, \theta) = (5, \pi)$ . We then ran (4) in the computational domain for 10,000 steps, returning the final position of the microswimmer, before returning its position in the computational domain, corresponding to a position and orientation in a physical domain. This process was carried out 5,000 times in order to generate the histograms 6a and 6b. We see that our microswimmers are overwhelmingly more likely to accumulate on

or near the boundaries of the vessel, and orientate parallel to them, as seen by the pronounced spikes at  $\theta = 0, \pi, 2\pi$  in Figure 6a.



(a) A histogram generated through Monte Carlo simulations showing the numerical solution to the stochastic hydrodynamic model (10a).

(b) A histogram generated through Monte Carlo simulations of equation (10b) of the stochastic hydrodynamic model.

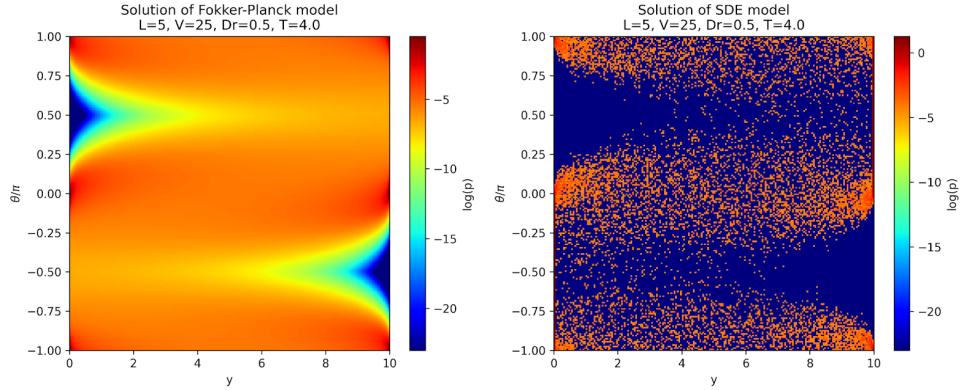


(c) A plot of the analytical expression (8), derived from the stresslet solution of a force dipole near a non-slip boundary.

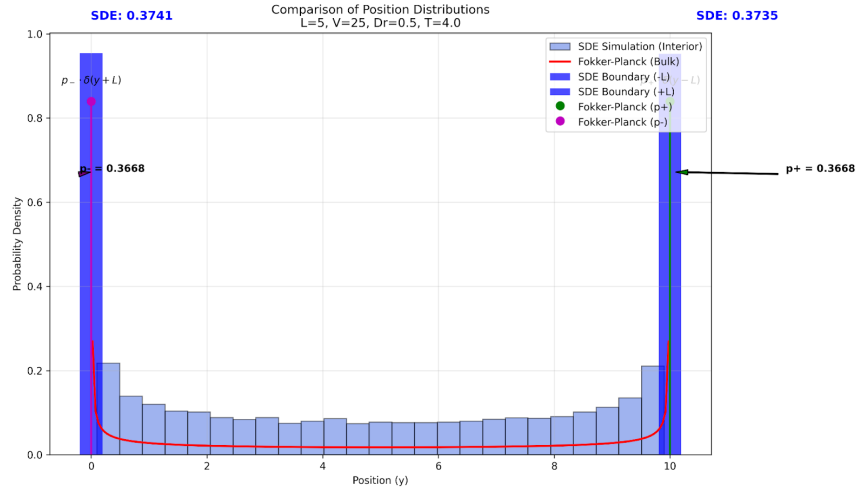
Figure 7: Three plots illustrating the boundary accumulation of microswimmers. Figure 7a and Figure 7b are derived from the stochastic equations considered in Model 2, whereas Figure 7c is derived from the analytical expression (8).

Model 2 shows qualitatively similar results to Model 1. In this case, as the stochastic simulations were more computationally intensive, we used 2,500 trajectories, with all other steps identical. Our far-field Fokker-Planck equation retains a similar profile to that shown in Figure 7a, and in particular to the ‘theory’ line in Figure 6c, from which this model was derived. We had previously described how expression (8) has used a far-field approximation to find an analytical solution to the Fokker-Planck

equation. We see this in Figure 7c by the singularities at  $y = 0$  and  $y = H$ . This is a weakness of this model, as it results in numerical instability, requiring the truncation of the domain (such as the computational domain  $y \in [1, 9]$  considered here) such that we can normalise our values to make a valid probability distribution plot that integrates to unity. Although this is a far-field approximation, we find as in Figure 6c that the expression does remarkably well modelling the distribution of microswimmers near to but not reaching the boundary.



(a) Two heat maps corresponding to the solution of the Fokker-Planck system (left) and the stochastic system (right) of Model 3, solved through finite difference schemes.



(b) A histogram of the  $y$ -distribution of swimmers in Model 3, with a colour legend to show the outputs of the individual degenerate components.

Figure 8: The generated heat maps and corresponding histogram of Model 3. Figure 8a via finite difference schemes and Figure 8b by Monte Carlo simulations.

For Figure 8a, we had used a finite difference scheme described in Fu *et al.* [5] and provided by the authors. Although small, there is a darkening of the boundaries in the left-hand image, corresponding to the same boundary accumulation we have come to expect from models 1 and 2, found at  $\theta = 0, \pi, 2\pi$ . The blue trian-

gles that protrude into the heat map correspond to boundary conditions (13b) and (13c), enforcing that microswimmers cannot enter the phases from outside the vessel. Figure 8b shows the histogram using Monte Carlo simulations largely the same as in the previous two models, where we have used different colours to denote the histogram bins corresponding to different phases, and similarly for the corresponding Fokker-Planck numerical solution in each phase.

Turning now to the shortcomings of our models, we shall begin by considering some shared limitations. Perhaps the most obvious problem with our models is their dimensions. Although we can generalise easily to the  $x$ -direction as previously stated, we cannot consider all three-dimensions. We therefore neglect the richer dynamics presented by physical reality. This is non-trivial, since we had previously stated that flagella usually move in a chiral motion [8], and as a result so may our microswimmer. Furthermore, we had considered microswimmers as perfectly rigid ellipses (where shape was relevant), but had assumed their propulsion is aligned with their geometric centre. In reality, a pusher would normally have its thrust applied behind its centre of mass and a puller in front of it. This is a drawback of our computational domain method. For each model, we had modelled one microswimmer per trajectory. In reality, we know from hydrodynamic-dipole interactions as mentioned in Lauga [8] and Koch and Subramanian [7] that microswimmers exhibit quantitatively different dynamics on small-scales when in higher cell densities compared to in individual cells. Depending on the influence of this additional phenomenon, this would represent a mechanical difference between the considerations of our models and our motivations from initial observations described by Rothschild [11] and Berke *et al.* [1].

Focusing now on Model 1, we note that it does not distinguish between pullers and pushers. This is important regarding Figure 6a, in particular the marginal in  $\theta$ . We see that it has the same shape as the histogram in Figure 7b. This is important as we had explicitly assumed that  $P > 0$ , thereby modelling pushers. As a result, Model 1 implicitly models pushers, and hence if we had assumed  $P < 0$ , this model would be inadequate, as pullers have a different probability distribution in  $\theta$  that we see in Figure 13 of Chen and Thiffeault [2]. By neglecting hydrodynamic terms, we fail to capture the rate at which microswimmers orient themselves with the boundary, in comparison to Model 2. This is because from Berke *et al.* [1], we see that a purely stochastic re-orientation would occur in around  $D_\theta^{-1} \sim 10^2$  s, in comparison to  $\Omega_z^{-1} \sim \mathcal{O}(1)$  s in Model 2. From physical observation, we know that cells re-orient themselves in a few seconds, showing that hydrodynamic terms dominate microswimmer re-orientation.

We find that most of the literature surrounding Model 2 has been used to model *E. coli*. In particular, the derivations as such have assumed that our microswimmers are *non-tumbling*, as stated in Berke *et al.* [1], where the authors have used mutant strain HCB-437, although we know from Lauga [8] that most microswimmers and strains of *E. coli* in general exhibit *tumbling*. This is when a microswimmer re-orient itself deterministically, by reversing the rotation direction from one of its flagella (see Figure 4a for reference) in order to change direction. The second shortcoming is the inaccuracy of the analytical section of Model 2, at the boundaries, resulting from the singularities of the stokeslet. This is particularly unfortunate, as we know from above that hydrodynamic terms dominate some aspects of the microswimmer-boundary interaction.

The main weakness of Model 3 is the lack of accounting for the shape of the microswimmer. We understand that this results from a desire to simplify boundary interactions, although this results in the later problem of non-trivial boundary conditions and their implementation. We know from Chen and Thiffeault [2] that the shape of the microswimmer plays a role in the accumulation profile, particularly regarding the  $\theta$  distribution of that profile. In an attempt to standardise our models, we had attempted to integrate ‘shape’ into this model by setting the boundary-contacting phase a distance  $\bar{y} = \frac{1}{2\pi} \int_0^{2\pi} \sqrt{a^2 \sin^2(\theta) + b^2 \cos^2(\theta)} d\theta$  from  $y = 0$  and  $y = H$ , where we have defined  $a$  and  $b$  as previously. The problem herein lies since this would hold for any shape of the same ‘average radius’ as our microswimmer.

## 4 Conclusion

In this Case Study, we have investigated the phenomenon of boundary accumulation of randomly moving microswimmers, and we have employed three models to do so. Model 1 was a purely stochastic model that incorporated the elliptical shape of the microswimmers to show their accumulation. Model 2 consisted of a purely hydrodynamic model which relied upon the stokeslet solution to Stokes’ equations for a microswimmer modelled as a force dipole, and a secondary model that used these hydrodynamic corrections applied to Model 1. Model 3 aimed to simplify the physical situation by removing the challenging near-boundary dynamics through distinguishing between microswimmers on the boundary and in the bulk of the fluid. This resulted in a degenerate Fokker-Planck system which did not explicitly include the shape of the microswimmer. Overall, all models broadly successful in illustrating the phenomenon of boundary accumulation, although stochastic Model 2 with hydrodynamic corrections had the added benefit of re-orientation accuracy.

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